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End Semester Examination 2025

Name of the Course: B. Tech, EE

Semester: 6th

Name of the Paper – Power System-II

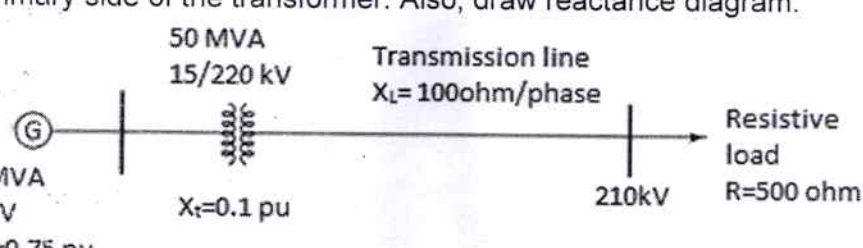
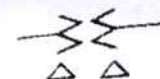
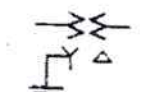
Paper Code: TEE 602

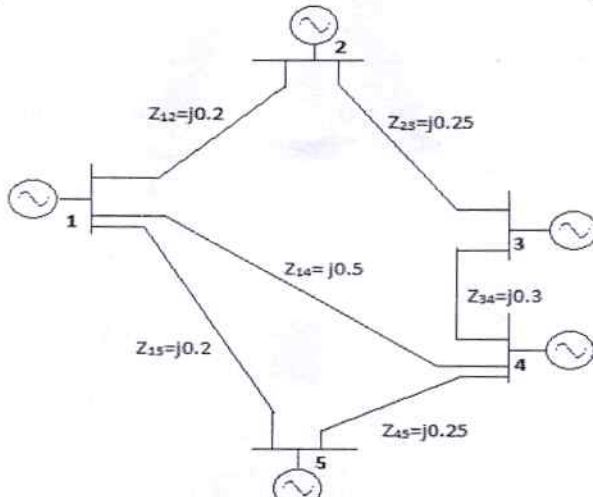
Time: 3Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Show that total reactance of the transformer (in per unit system) referred to primary and referred to secondary are same. Also, discuss the relation between line and phase voltage in p.u for the star and delta connected load.	
(b)	The one-line diagram for three generator systems is shown in Figure. Determine reactance of each component on base 75 MVA and 11kV for the primary side of the transformer. Also, draw reactance diagram. 	CO1
(c)	A 3-phase transformer rated 100 MVA, 220/22 kV has a leakage impedance of 8%. Calculate the per unit impedance referred to the low-voltage side and high-voltage side. Also, explain the impact of using different base values for MVA and voltage levels.	
Q2	(20 marks)	
(a)	Draw zero sequence network for the following transformers: (i)  (ii)  (iii) 	CO2
(b)	In a three-phase four wire system the currents in the lines a, b, and c under abnormal conditions of loading were as follows; $I_a = 50 \angle 45^\circ, I_b = 40 \angle 300^\circ, I_c = 30 \angle 180^\circ$ Calculate the zero, positive, and negative phase sequence currents in line a and the return current in the neutral conductor.	

(c)	Derive the complex power in terms of symmetrical components.	
Q3	(20 marks)	CO3
(a)	Show that under the condition of LG faults (i) all sequence networks are existing (ii) all sequence networks are interconnected in series.	
(b)	A single line-to-ground fault occurs on phase A of an unloaded 11 kV generator. The positive, negative, and zero sequence impedances of the generator are: $Z_1=j0.2$ p.u., $Z_2=j0.2$ p.u., and $Z_0=j0.05$ p.u. Assume pre-fault Calculate: - The fault current in per unit - The sequence components of fault current	
(c)	A synchronous generator rated 500 KVA, 440 V, 0.1 p.u. subtransient reactance is supplying a passive load of 400 kW at 0.8 lag p.f. Calculate the initial symmetrical r.m.s. current for a 3-phase fault at generator terminals.	
Q4	(20 marks)	CO4
(a)	Discuss the Gauss-Seidel method of load flow analysis	
(b)	Determine Y_{bus} for the 5-bus system as shown in figure below. The line series impedances between the busses are indicated in figure 	
(c)	Derive static load flow equations and discuss the desirable feature of an ideal load flow analysis	
Q5	(20 marks)	CO5, CO6
(a)	What do you mean by power system stability. Discuss the types of stability existing in power systems	
(b)	A synchronous generator is supplying power to infinite bus through a transmission line. Derive the expression for stability limit in terms of A , B , C , and D constant of transmission line. Assume all the essential data which are required.	
(c)	The A, B, C and D constants of the nominal π -network representing a three-phase transmission line are: $A = D = 0.97 \angle 0.6^\circ$, $B = 60 \angle 70^\circ \Omega$ and $C = 0.001 \angle 91^\circ S$. Find the steady state stability limit if both the sending and receiving end voltages are at 132 kV (a) with the ABCD constants as given (b) with the shunt admittance neglected.	